

Hilbert Space Theory: Fundamental Questions and Answers

Abstract

This document provides comprehensive answers to fundamental questions about Hilbert space theory, the mathematical foundation of quantum mechanics. Each question is followed by detailed explanations, definitions, examples, and key properties.

Introduction

The theory of Hilbert spaces provides the rigorous mathematical framework for quantum mechanics. Understanding these concepts is essential for mastering quantum theory and its applications in modern physics.

1 Complex Vector Spaces

Question (a)

What is a complex vector space?

Definition 1 (Complex Vector Space). A **complex vector space** V over the field of complex numbers $\mathbb{C}$ is a set equipped with two operations:

- **Vector addition:** $+: V \times V \rightarrow V$
- **Scalar multiplication:** $\cdot: \mathbb{C} \times V \rightarrow V$

satisfying the following axioms for all $|u\rangle, |v\rangle, |w\rangle \in V$ and $\alpha, \beta \in \mathbb{C}$:

Addition Axioms:

1. **Commutativity:** $|u\rangle + |v\rangle = |v\rangle + |u\rangle$
2. **Associativity:** $(|u\rangle + |v\rangle) + |w\rangle = |u\rangle + (|v\rangle + |w\rangle)$
3. **Identity element:** $\exists |0\rangle \in V$ such that $|v\rangle + |0\rangle = |v\rangle$
4. **Inverse elements:** $\forall |v\rangle \in V, \exists -|v\rangle$ such that $|v\rangle + (-|v\rangle) = |0\rangle$

Scalar Multiplication Axioms:

5. **Compatibility:** $\alpha(\beta|v\rangle) = (\alpha\beta)|v\rangle$
6. **Identity:** $1|v\rangle = |v\rangle$
7. **Distributivity (vectors):** $\alpha(|u\rangle + |v\rangle) = \alpha|u\rangle + \alpha|v\rangle$

8. **Distributivity (scalars):** $(\alpha + \beta) |v\rangle = \alpha |v\rangle + \beta |v\rangle$

Example 1 (Common Complex Vector Spaces). • $\mathbb{C}^n$: n -tuples of complex numbers

- Space of complex-valued continuous functions on $[a, b]$
- Space of square-integrable functions $L^2(\mathbb{R})$
- Space of quantum states in quantum mechanics

Key Point: Unlike real vector spaces, complex vector spaces allow scalar multiplication by complex numbers, which is essential for quantum mechanics where phases (complex numbers of unit modulus) carry physical significance.

2 Inner and Outer Products

Question (b)

What's an inner product? and an outer product?

Inner Product

Definition 2 (Inner Product). An **inner product** on a complex vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ satisfying:

1. **Conjugate symmetry:** $\langle v|u\rangle = \langle u|v\rangle^*$
2. **Linearity in second argument:**

$$\langle u|\alpha v + \beta w\rangle = \alpha \langle u|v\rangle + \beta \langle u|w\rangle \quad (1)$$

3. **Positive definiteness:**

$$\langle v|v\rangle \geq 0 \quad \text{and} \quad \langle v|v\rangle = 0 \iff |v\rangle = |0\rangle \quad (2)$$

Note: By conjugate symmetry, the inner product is *antilinear* (conjugate linear) in the first argument:

$$\langle \alpha u + \beta v|w\rangle = \alpha^* \langle u|w\rangle + \beta^* \langle v|w\rangle \quad (3)$$

Physical Interpretation: In quantum mechanics, $|\langle \psi|\phi\rangle|^2$ represents the probability amplitude that a system in state $|\phi\rangle$ will be found in state $|\psi\rangle$ upon measurement.

Example 2 (Standard Inner Products). • **Finite dimensions** ($\mathbb{C}^n$):

$$\langle u|v\rangle = \sum_{i=1}^n u_i^* v_i \quad (4)$$

- **Function space** ($L^2(\mathbb{R})$):

$$\langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x)g(x) dx \quad (5)$$

Outer Product

Definition 3 (Outer Product). The **outer product** of two vectors $|v\rangle \in V$ and $\langle u| \in V^*$ (dual space) is the operator:

$$|v\rangle \langle u| : V \rightarrow V \quad (6)$$

defined by its action on any vector $|w\rangle \in V$:

$$(|v\rangle \langle u|) |w\rangle = |v\rangle \langle u|w\rangle = \langle u|w\rangle |v\rangle \quad (7)$$

Key Properties:

- The outer product is a *linear operator* (rank-1 operator)
- In matrix form (finite dimensions):

$$(|v\rangle \langle u|)_{ij} = v_i u_j^* \quad (8)$$

- The outer product is *not commutative*: $|v\rangle \langle u| \neq |u\rangle \langle v|$

Special Case - Projection Operators:

$$P_v = |v\rangle \langle v| \quad (\text{for normalized } |v\rangle) \quad (9)$$

projects any vector onto the subspace spanned by $|v\rangle$.

Example 3 (Outer Product in $\mathbb{C}^2$). Let $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Then:

$$|0\rangle \langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (10)$$

Completeness Relation: For an orthonormal basis $\{|e_i\rangle\}$:

$$\sum_i |e_i\rangle \langle e_i| = \mathbb{I} \quad (11)$$

This is fundamental for expanding operators and states.

3 Kets and Bras

Question (c)

What's a *ket*? and a *bra*?

Dirac Notation

The **bra-ket notation**, introduced by Paul Dirac, is the standard notation in quantum mechanics.

Definition 4 (Ket). A **ket** $|\psi\rangle$ is a vector in the Hilbert space $\mathcal{H}$. It represents a quantum state.

Notation: The symbol $|\cdot\rangle$ is read as "ket"

Properties:

- *Linear combination:* $|\psi\rangle = \alpha|\phi_1\rangle + \beta|\phi_2\rangle$ where $\alpha, \beta \in \mathbb{C}$
- *Normalization:* $\langle\psi|\psi\rangle = 1$ for normalized states
- *Physical meaning:* Complete description of quantum state

Definition 5 (Bra). A **bra** $\langle\psi|$ is a vector in the dual space $\mathcal{H}^*$. It is the conjugate transpose of the corresponding ket.

Notation: The symbol $\langle\cdot|$ is read as "bra"

Relationship to ket:

$$\langle\psi| = (|\psi\rangle)^\dagger \quad (12)$$

where $\dagger$ denotes the Hermitian conjugate (conjugate transpose).

Mathematical Connection

In finite dimensions:

$$|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \iff \langle\psi| = (c_1^* \ c_2^* \ \cdots \ c_n^*) \quad (13)$$

In function space:

$$|\psi\rangle \equiv \psi(x) \iff \langle\psi| \equiv \psi^*(x) \quad (14)$$

Operations with Bras and Kets

1. **Inner product (bra-ket):**

$$\langle\phi|\psi\rangle = \text{complex number} \quad (15)$$

Closing the "bra-ket" gives a scalar.

2. **Outer product (ket-bra):**

$$|\psi\rangle\langle\phi| = \text{operator} \quad (16)$$

3. **Operator action:**

$$\hat{A}|\psi\rangle = |\psi'\rangle \quad (\text{ket}) \quad (17)$$

$$\langle\psi|\hat{A} = \langle\psi'| \quad (\text{bra}) \quad (18)$$

4. **Matrix element:**

$$\langle\phi|\hat{A}|\psi\rangle = \text{complex number} \quad (19)$$

Example 4 (Spin-1/2 States). For a spin-1/2 particle:

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \langle\uparrow| = (1 \ 0) \quad (20)$$

$$|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \langle\downarrow| = (0 \ 1) \quad (21)$$

Inner product: $\langle\uparrow|\downarrow\rangle = 0$ (orthogonal states)

Superposition: $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$

Etymology: "Bra-ket" comes from splitting the word "bracket": $\underbrace{\langle\cdot|}_{\text{bra}} \underbrace{|\cdot\rangle}_{\text{ket}}$

4 Euclidean vs Hilbert Spaces

Question (d)

What's a Euclidean space? Is it the same as a Hilbert space?

Definition 6 (Euclidean Space). A **Euclidean space** $\mathbb{R}^n$ is a finite-dimensional real vector space with the standard inner product:

$$\langle \vec{u}, \vec{v} \rangle = \sum_{i=1}^n u_i v_i \quad (22)$$

which induces the Euclidean norm (distance):

$$\|\vec{v}\| = \sqrt{\sum_{i=1}^n v_i^2} \quad (23)$$

Definition 7 (Hilbert Space). A **Hilbert space** $\mathcal{H}$ is a complete inner product space (over $\mathbb{C}$ or $\mathbb{R}$) with the following properties:

1. It is a vector space (over $\mathbb{C}$ in QM)
2. It has an inner product $\langle \cdot | \cdot \rangle$
3. It is **complete**: every Cauchy sequence converges to a limit in $\mathcal{H}$
4. It can be finite or infinite dimensional

Key Differences

Property	Euclidean Space	Hilbert Space
Field	Real ($\mathbb{R}$)	Complex ($\mathbb{C}$) typically
Dimension	Finite ($n < \infty$)	Finite or infinite
Completeness	Always complete	Always complete
Inner product	Real-valued	Complex-valued
Examples	$\mathbb{R}^2, \mathbb{R}^3$	$\mathbb{C}^n, L^2(\mathbb{R}), \ell^2$

Relationship

Euclidean space as a Hilbert space:

- Every finite-dimensional Euclidean space $\mathbb{R}^n$ is a (real) Hilbert space
- It's complete (all Cauchy sequences converge)
- However, quantum mechanics requires *complex* Hilbert spaces

Key Statement:

A Euclidean space is a special case of a Hilbert space (finite-dimensional, real). However, in quantum mechanics, we use *complex* Hilbert spaces, which can be infinite-dimensional. So while related, they are not identical.

Example 5 (Comparison). • **Euclidean:** 3D position space $\mathbb{R}^3$ with $\vec{r} = (x, y, z)$

- **Hilbert:** Quantum state space $L^2(\mathbb{R}^3)$ with $\psi(\vec{r})$

The quantum wavefunction lives in an infinite-dimensional complex Hilbert space, not a Euclidean space.

Geometric Intuition

Both spaces share geometric properties:

- Orthogonality: $\langle u | v \rangle = 0$
- Pythagorean theorem: $\|u + v\|^2 = \|u\|^2 + \|v\|^2$ (if orthogonal)
- Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$

However, Hilbert spaces extend these concepts to infinite dimensions with complex scalars.

5 Linear Operators

Question (e)

What's a linear operator? Enumerate the most important types of operators and their properties? How about the most important operations between or on them?

Definition 8 (Linear Operator). A **linear operator** $\hat{A} : \mathcal{H} \rightarrow \mathcal{H}$ is a map between Hilbert spaces satisfying:

$$\hat{A}(\alpha |\psi\rangle + \beta |\phi\rangle) = \alpha \hat{A} |\psi\rangle + \beta \hat{A} |\phi\rangle \quad (24)$$

for all $|\psi\rangle, |\phi\rangle \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{C}$.

Most Important Types of Operators

1. Hermitian (Self-Adjoint) Operators

Definition 9 (Hermitian Operator). An operator $\hat{A}$ is **Hermitian** if:

$$\hat{A} = \hat{A}^\dagger \iff \langle \phi | \hat{A} \psi \rangle = \langle \hat{A} \phi | \psi \rangle \quad (25)$$

Properties:

- Eigenvalues are *real*: $\hat{A} |a\rangle = a |a\rangle \Rightarrow a \in \mathbb{R}$
- Eigenvectors with different eigenvalues are orthogonal
- Spectral theorem: Can be diagonalized with orthonormal eigenbasis
- **Physical significance:** Represent observable quantities (position, momentum, energy, etc.)

Example 6. Position operator: $\hat{x}$, Momentum operator: $\hat{p}$, Hamiltonian: $\hat{H}$

2. Unitary Operators

Definition 10 (Unitary Operator). An operator $\hat{U}$ is **unitary** if:

$$\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \mathbb{I} \quad (26)$$

Properties:

- Preserves inner products: $\langle \hat{U}\phi | \hat{U}\psi \rangle = \langle \phi | \psi \rangle$
- Preserves norms: $\| \hat{U} |\psi\rangle \| = \| |\psi\rangle \|$
- Eigenvalues have unit modulus: $|u| = 1$
- **Physical significance:** Represent time evolution and symmetry transformations

Example 7. Time evolution operator: $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$
Rotation operators, translation operators

3. Projection Operators

Definition 11 (Projection Operator). An operator $\hat{P}$ is a **projection** if:

$$\hat{P}^2 = \hat{P} \quad \text{and} \quad \hat{P}^\dagger = \hat{P} \quad (27)$$

Properties:

- Eigenvalues are 0 or 1
- Projects onto a subspace: $\hat{P} |\psi\rangle$ is the component of $|\psi\rangle$ in that subspace
- **Physical significance:** Represent measurement processes

Example 8. For orthonormal basis $\{|n\rangle\}$:

$$\hat{P}_n = |n\rangle \langle n| \quad (28)$$

Completeness: $\sum_n \hat{P}_n = \mathbb{I}$

4. Normal Operators

Definition 12 (Normal Operator). An operator $\hat{A}$ is **normal** if:

$$[\hat{A}, \hat{A}^\dagger] = \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} = 0 \quad (29)$$

Note: Hermitian and unitary operators are special cases of normal operators.

5. Positive Operators

Definition 13 (Positive Operator). An operator $\hat{A}$ is **positive** (or positive semi-definite) if:

$$\langle \psi | \hat{A} \psi \rangle \geq 0 \quad \forall |\psi\rangle \quad (30)$$

Properties:

- All eigenvalues are non-negative
- Can be written as $\hat{A} = \hat{B}^\dagger \hat{B}$ for some operator $\hat{B}$
- **Physical significance:** Density matrices are positive operators

6. Commuting Operators

Two operators $\hat{A}$ and $\hat{B}$ **commute** if:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = 0 \quad (31)$$

Properties:

- Share a complete set of eigenvectors
- Can be simultaneously diagonalized
- **Physical significance:** Commuting observables can be measured simultaneously with arbitrary precision

Example 9. $[\hat{x}, \hat{p}] = i\hbar$ (do not commute - uncertainty principle)
 $[\hat{L}^2, \hat{L}_z] = 0$ (commute - quantum numbers ℓ, m)

Important Operations on Operators

1. Hermitian Conjugate (Adjoint)

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A}^\dagger \phi | \psi \rangle \quad (32)$$

In matrix form: $(\hat{A}^\dagger)_{ij} = (\hat{A}_{ji})^*$ (conjugate transpose)

2. Commutator

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \quad (33)$$

Properties:

- Anticommutative: $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$
- Jacobi identity: $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$
- Leibniz rule: $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$

3. Anticommutator

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A} \quad (34)$$

Important in fermionic systems.

4. Function of Operators

For Hermitian $\hat{A}$ with spectral decomposition $\hat{A} = \sum_n a_n |n\rangle \langle n|$:

$$f(\hat{A}) = \sum_n f(a_n) |n\rangle \langle n| \quad (35)$$

Examples:

- Exponential: $e^{\hat{A}} = \sum_{n=0}^{\infty} \frac{\hat{A}^n}{n!}$
- Time evolution: $e^{-i\hat{H}t/\hbar}$

5. Trace

$$\text{Tr}(\hat{A}) = \sum_n \langle n | \hat{A} | n \rangle \quad (36)$$

Properties:

- Cyclic: $\text{Tr}(\hat{A}\hat{B}) = \text{Tr}(\hat{B}\hat{A})$
- Linear: $\text{Tr}(\alpha\hat{A} + \beta\hat{B}) = \alpha\text{Tr}(\hat{A}) + \beta\text{Tr}(\hat{B})$
- Basis-independent

6. Tensor Product

For operators $\hat{A} \in \mathcal{H}_1$ and $\hat{B} \in \mathcal{H}_2$:

$$(\hat{A} \otimes \hat{B})(|\psi\rangle \otimes |\phi\rangle) = (\hat{A}|\psi\rangle) \otimes (\hat{B}|\phi\rangle) \quad (37)$$

Essential for composite systems.

Key Theorems

Theorem 1 (Spectral Theorem). *Every Hermitian operator on a finite-dimensional Hilbert space can be diagonalized:*

$$\hat{A} = \sum_n a_n |a_n\rangle \langle a_n| \quad (38)$$

where $a_n \in \mathbb{R}$ and $\{|a_n\rangle\}$ form an orthonormal basis.

Theorem 2 (Uncertainty Principle). *For non-commuting Hermitian operators $\hat{A}$ and $\hat{B}$:*

$$\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle| \quad (39)$$

6 Basis of Hilbert Space

Question (f)

What's a basis of a Hilbert space? Is it unique?

Definition 14 (Basis of Hilbert Space). *A **basis** (or complete orthonormal system) of a Hilbert space $\mathcal{H}$ is a set of vectors $\{|e_i\rangle\}_{i \in I}$ such that:*

1. Orthonormality:

$$\langle e_i | e_j \rangle = \delta_{ij} \quad (40)$$

2. Completeness: Every vector $|\psi\rangle \in \mathcal{H}$ can be uniquely expressed as:

$$|\psi\rangle = \sum_{i \in I} c_i |e_i\rangle \quad \text{where } c_i = \langle e_i | \psi \rangle \quad (41)$$

For infinite-dimensional spaces, this sum converges in the Hilbert space norm.

Completeness Relation

The completeness of a basis is expressed by the identity:

$$\boxed{\sum_{i \in I} |e_i\rangle \langle e_i| = \mathbb{I}} \quad (42)$$

This is also called the **resolution of identity** or **closure relation**.

Is the Basis Unique?

Answer: No, the basis is NOT unique.

A Hilbert space has *infinitely many* orthonormal bases. Any unitary transformation of one basis yields another valid basis.

Theorem 3 (Non-uniqueness of Basis). *If $\{|e_i\rangle\}$ is an orthonormal basis and $\hat{U}$ is a unitary operator, then:*

$$\{|e'_i\rangle = \hat{U} |e_i\rangle\} \quad (43)$$

is also an orthonormal basis.

Example 10 (Multiple Bases for $\mathbb{C}^2$). **Computational basis:**

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (44)$$

Hadamard basis:

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (45)$$

Circular basis:

$$|R\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |L\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (46)$$

All three are valid orthonormal bases for the same space!

Change of Basis

Given two bases $\{|e_i\rangle\}$ and $\{|f_j\rangle\}$, we can expand one in terms of the other:

$$|f_j\rangle = \sum_i U_{ij} |e_i\rangle \quad (47)$$

where $U_{ij} = \langle e_i | f_j \rangle$ form a unitary matrix.

Vector transformation:

$$|\psi\rangle = \sum_i c_i |e_i\rangle = \sum_j c'_j |f_j\rangle \quad (48)$$

with $c'_j = \sum_i U_{ji}^* c_i$

Common Bases in Quantum Mechanics

1. **Position basis:** $\{|x\rangle\}$ with $\hat{x}|x\rangle = x|x\rangle$
2. **Momentum basis:** $\{|p\rangle\}$ with $\hat{p}|p\rangle = p|p\rangle$
3. **Energy basis:** $\{|E_n\rangle\}$ with $\hat{H}|E_n\rangle = E_n|E_n\rangle$
4. **Angular momentum basis:** $\{|\ell, m\rangle\}$

Physical Interpretation: Different bases correspond to different complete sets of compatible observables. The choice of basis depends on which observables are being measured or which provide the simplest description of the system.

Dimension and Cardinality

Finite dimension: A basis has n elements ($n < \infty$)

Infinite dimension (separable): A basis has countably infinite elements (like $\mathbb{N}$)

Non-separable: A basis has uncountably infinite elements (like $\mathbb{R}$)

Most physical Hilbert spaces are separable.

7 Infinite-Dimensional Hilbert Spaces

Question (g)

Given an infinite-dimensional Euclidean space, what's the defining property of the vectors contained in the corresponding Hilbert space? Are the remaining vectors not contained in the Hilbert space still useful? Can they form a basis of the Hilbert space? What's a continuous representation?

This question addresses subtle but crucial aspects of infinite-dimensional quantum mechanics.

Defining Property: Square-Integrability

Definition 15 (Square-Integrable Functions). *Given an infinite-dimensional space, the corresponding Hilbert space consists of vectors (functions) ψ such that:*

$$\|\psi\|^2 = \int |\psi(x)|^2 dx < \infty \quad (49)$$

This is the space $L^2(\mathbb{R})$ (or $L^2(\mathbb{R}^n)$ for higher dimensions).

Key Point: The inner product must be finite:

$$\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty \quad (50)$$

This ensures:

- The wavefunction is normalizable (can represent a probability density)
- The state has finite energy
- Mathematical operations are well-defined

Vectors NOT in the Hilbert Space

Many important objects in quantum mechanics are *not* square-integrable:

Example 11 (Non-Hilbert Space Vectors). 1. **Position eigenstates:** $|x_0\rangle$ where $\psi(x) = \delta(x - x_0)$

$$\int |\delta(x - x_0)|^2 dx = \infty \quad (51)$$

2. **Momentum eigenstates:** $|p_0\rangle$ where $\psi(x) = e^{ipx/\hbar}$

$$\int |e^{ipx/\hbar}|^2 dx = \int dx = \infty \quad (52)$$

3. **Energy eigenstates in continuum:** Free particle states with $E > 0$

Are These "Non-Hilbert" Vectors Useful?

Answer: Absolutely! They are essential.

Despite not being in the Hilbert space, these generalized vectors (called **distributions** or **generalized functions**) are indispensable because:

1. **They diagonalize important observables:**

- $\hat{x}|x\rangle = x|x\rangle$ (position)
- $\hat{p}|p\rangle = p|p\rangle$ (momentum)

2. **They form a "generalized basis":** Any physical state can be expanded:

$$|\psi\rangle = \int dx \psi(x) |x\rangle = \int dp \tilde{\psi}(p) |p\rangle \quad (53)$$

3. **They simplify calculations:** Position and momentum representations are standard tools

4. **They have clear physical meaning:** Represent idealized measurement outcomes

Can They Form a Basis?

Answer: Yes, but as a "continuous basis" or "generalized basis"

Position eigenstates $\{|x\rangle\}_{x \in \mathbb{R}}$ satisfy:

1. **Orthonormality (generalized):**

$$\langle x|x'\rangle = \delta(x - x') \quad (54)$$

2. **Completeness (generalized):**

$$\int_{-\infty}^{\infty} dx |x\rangle \langle x| = \mathbb{I} \quad (55)$$

This is a **continuous basis** (uncountable infinity) vs. a discrete basis (countable infinity).

Rigged Hilbert Space

The proper mathematical framework is the **rigged Hilbert space** (or Gelfand triple):

$$\Phi \subset \mathcal{H} \subset \Phi^* \quad (56)$$

where:

- Φ : Space of "nice" test functions (rapidly decreasing)
- $\mathcal{H}$: Hilbert space $L^2(\mathbb{R})$
- Φ^* : Space of distributions (generalized functions)

Key insight: Eigenstates like $|x\rangle$ and $|p\rangle$ live in Φ^* , not $\mathcal{H}$.

Continuous Representation

Definition 16 (Continuous Representation). A **continuous representation** uses continuous basis states (like position or momentum) to represent quantum states:

Position representation:

$$\psi(x) = \langle x|\psi\rangle \quad (57)$$

Momentum representation:

$$\tilde{\psi}(p) = \langle p|\psi\rangle \quad (58)$$

These are related by the Fourier transform:

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar} \psi(x) dx \quad (59)$$

Practical Consequences

1. **Normalization:** Physical states must be normalized:

$$\int |\psi(x)|^2 dx = 1 \quad (60)$$

2. **Wave packets:** Represent particles with localized, square-integrable functions
3. **Scattering theory:** Use non-normalizable plane waves, but calculate normalizable cross-sections
4. **Delta-function normalization:** For continuum states:

$$\langle E|E'\rangle = \delta(E - E') \quad (61)$$

Bottom line: The "proper" Hilbert space vectors are square-integrable, but the "improper" generalized eigenstates are indispensable tools that extend the framework in a mathematically rigorous way (via distribution theory).

8 $\ell^2(\infty)$ and $L^2(x)$ Spaces

Question (h)

Do you understand the most characteristic examples of infinite-dimensional spaces, denoted by $\ell^2(\infty)$ and $L^2(x)$ spaces? What do we mean when we say that they are isomorphic to each other? And when we say that *all* infinite-dimensional Hilbert spaces are isomorphic to them?

The Space ℓ^2 (little ell-two)

Definition 17 (ℓ^2 Space). The space ℓ^2 (or $\ell^2(\mathbb{N})$ or $\ell^2(\infty)$) consists of all infinite sequences of complex numbers $\{c_n\}_{n=1}^{\infty}$ such that:

$$\sum_{n=1}^{\infty} |c_n|^2 < \infty \quad (62)$$

Inner product:

$$\langle \{a_n\}, \{b_n\} \rangle = \sum_{n=1}^{\infty} a_n^* b_n \quad (63)$$

Norm:

$$\|\{c_n\}\| = \sqrt{\sum_{n=1}^{\infty} |c_n|^2} \quad (64)$$

Example 12 (ℓ^2 sequences). • $\{1, 1/2, 1/3, 1/4, \dots\} \in \ell^2$ since $\sum 1/n^2 = \pi^2/6$

- $\{1, 1, 1, 1, \dots\} \notin \ell^2$ since $\sum 1 = \infty$
- $\{1, 0, 0, 0, \dots\}$ (standard basis vector) $\in \ell^2$

Physical example: Discrete energy levels of bound states:

$$|\psi\rangle = \sum_{n=1}^{\infty} c_n |n\rangle \quad (65)$$

The coefficients $\{c_n\}$ form an ℓ^2 sequence.

The Space $L^2(\mathbb{R})$ (big L-two)

Definition 18 (L^2 Space). The space $L^2(\mathbb{R})$ (or $L^2(x)$) consists of all measurable functions $f : \mathbb{R} \rightarrow \mathbb{C}$ such that:

$$\int_{-\infty}^{\infty} |f(x)|^2 dx < \infty \quad (66)$$

Inner product:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f^*(x) g(x) dx \quad (67)$$

Norm:

$$\|f\| = \sqrt{\int_{-\infty}^{\infty} |f(x)|^2 dx} \quad (68)$$

Example 13 (L^2 functions). • $f(x) = e^{-x^2/2} \in L^2(\mathbb{R})$ (Gaussian)

- $f(x) = \frac{1}{1+x^2} \in L^2(\mathbb{R})$ (Lorentzian)
- $f(x) = 1 \notin L^2(\mathbb{R})$ (constant function)
- $f(x) = e^{ikx} \notin L^2(\mathbb{R})$ (plane wave)

Physical example: Quantum wavefunctions in position representation:

$$\psi(x) \in L^2(\mathbb{R}) \quad (69)$$

Isomorphism Between ℓ^2 and L^2

Definition 19 (Isomorphism of Hilbert Spaces). *Two Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ are **isomorphic** if there exists a bijective linear map $T : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ that preserves inner products:*

$$\langle T(\psi), T(\phi) \rangle_{\mathcal{H}_2} = \langle \psi, \phi \rangle_{\mathcal{H}_1} \quad (70)$$

*Such a map is called a **unitary isomorphism** or **Hilbert space isomorphism**.*

What does isomorphism mean?

- The spaces have the "same structure"
- They are "equivalent" from a Hilbert space perspective
- All geometric and algebraic properties are preserved
- They differ only in "labeling" or "representation"

Why are ℓ^2 and $L^2(\mathbb{R})$ Isomorphic?

Both are **separable infinite-dimensional** Hilbert spaces.

Key fact: They both have a countable orthonormal basis.

For ℓ^2 : Standard basis:

$$e_n = (0, 0, \dots, 0, \underbrace{1}_{n\text{-th position}}, 0, \dots) \quad (71)$$

For $L^2(\mathbb{R})$: Hermite functions form a complete orthonormal basis:

$$\phi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} e^{-x^2/2} H_n(x) \quad (72)$$

Isomorphism construction:

$$T : \ell^2 \rightarrow L^2(\mathbb{R}) \quad (73)$$

$$T(\{c_n\}) = \sum_{n=0}^{\infty} c_n \phi_n(x) \quad (74)$$

This map is:

- **Linear:** $T(\alpha\{c_n\} + \beta\{d_n\}) = \alpha T(\{c_n\}) + \beta T(\{d_n\})$
- **Bijective:** One-to-one and onto
- **Isometric:** $\|T(\{c_n\})\|_{L^2} = \|\{c_n\}\|_{\ell^2}$

Universal Property: All Separable Hilbert Spaces are Isomorphic

Theorem 4 (Classification of Separable Hilbert Spaces). *Any two infinite-dimensional separable Hilbert spaces are isomorphic to each other, and hence to ℓ^2 and $L^2(\mathbb{R})$.*

What does this mean?

Up to isomorphism, there is essentially **only one** infinite-dimensional separable Hilbert space. The differences between various realizations (sequences, functions on $\mathbb{R}$, functions on $[0, 1]$, etc.) are merely differences in representation.

Why is this true?

The key property is having a **countable orthonormal basis**. If two Hilbert spaces both have countable orthonormal bases:

$$\mathcal{H}_1 : \{|e_n\rangle\}, \quad \mathcal{H}_2 : \{|f_n\rangle\} \quad (75)$$

then the map $|e_n\rangle \mapsto |f_n\rangle$ extends to an isomorphism.

Examples of Isomorphic Spaces

All of the following are isomorphic (as Hilbert spaces):

1. $\ell^2(\mathbb{N})$ - square-summable sequences
2. $L^2(\mathbb{R})$ - square-integrable functions on the real line
3. $L^2([0, 1])$ - square-integrable functions on $[0, 1]$
4. $L^2(\mathbb{R}^3)$ - quantum wavefunctions in 3D
5. Space of square-integrable functions on any separable measure space

Practical Implications

1. **Representation freedom:** We can work in whichever representation is most convenient (discrete vs. continuous)
2. **Fourier analysis:** Transform between position (L^2) and momentum (L^2) - both are "the same" space
3. **Abstract theory:** Theorems proven for ℓ^2 apply to all separable Hilbert spaces
4. **Numerical methods:** Can discretize continuous problems (replace L^2 with ℓ^2 approximation)

Non-Separable Spaces

Important caveat: This isomorphism theorem applies only to **separable** Hilbert spaces (those with countable orthonormal bases).

Non-separable spaces (like $L^2(\mathbb{R}^\infty)$ for certain measures) can have different cardinalities and are not isomorphic to ℓ^2 .

Physical Interpretation

In quantum mechanics:

- **Discrete spectrum** (bound states): naturally described by ℓ^2
- **Continuous spectrum** (scattering states): naturally described by L^2
- The isomorphism tells us these are different views of the same mathematical structure

Bottom line: The statement "all infinite-dimensional separable Hilbert spaces are isomorphic" means that abstractly, they are all "the same space" - they differ only in how we label vectors and represent them concretely. This is one of the most profound and useful results in the mathematics of quantum mechanics.

9 Tensor Products

Question (i)

What's a tensor product?

Definition 20 (Tensor Product of Hilbert Spaces). *The **tensor product** $\mathcal{H}_1 \otimes \mathcal{H}_2$ of two Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ is a new Hilbert space that describes composite quantum systems.*

For states $|\psi\rangle \in \mathcal{H}_1$ and $|\phi\rangle \in \mathcal{H}_2$, the tensor product state is:

$$|\psi\rangle \otimes |\phi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2 \quad (76)$$

Often written as: $|\psi\rangle |\phi\rangle$ or $|\psi, \phi\rangle$ or $|\psi\phi\rangle$

Construction and Properties

1. Bilinearity:

$$(\alpha |\psi_1\rangle + \beta |\psi_2\rangle) \otimes |\phi\rangle = \alpha (|\psi_1\rangle \otimes |\phi\rangle) + \beta (|\psi_2\rangle \otimes |\phi\rangle) \quad (77)$$

$$|\psi\rangle \otimes (\alpha |\phi_1\rangle + \beta |\phi_2\rangle) = \alpha (|\psi\rangle \otimes |\phi_1\rangle) + \beta (|\psi\rangle \otimes |\phi_2\rangle) \quad (78)$$

2. General state in tensor product space: Not all states are simple tensor products! The general state is:

$$|\Psi\rangle = \sum_{i,j} c_{ij} |e_i\rangle \otimes |f_j\rangle \quad (79)$$

If $|\Psi\rangle$ cannot be written as $|\psi\rangle \otimes |\phi\rangle$ for any single pair, it is **entangled**.

3. Inner product:

$$(\langle\psi_1| \otimes \langle\phi_1|)(|\psi_2\rangle \otimes |\phi_2\rangle) = \langle\psi_1|\psi_2\rangle \langle\phi_1|\phi_2\rangle \quad (80)$$

4. Dimension:

$$\dim(\mathcal{H}_1 \otimes \mathcal{H}_2) = \dim(\mathcal{H}_1) \times \dim(\mathcal{H}_2) \quad (81)$$

Basis of Tensor Product Space

If $\{|e_i\rangle\}$ is a basis for $\mathcal{H}_1$ and $\{|f_j\rangle\}$ is a basis for $\mathcal{H}_2$, then:

$$\{|e_i\rangle \otimes |f_j\rangle\}_{i,j} \quad (82)$$

is a basis for $\mathcal{H}_1 \otimes \mathcal{H}_2$.

Example 14 (Two Qubits). *Each qubit has basis $\{|0\rangle, |1\rangle\}$ with $\dim = 2$.*

Two-qubit system: $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$

Basis: $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

General state:

$$|\Psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle \quad (83)$$

Entangled state example (Bell state):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (84)$$

This cannot be written as $|\psi\rangle \otimes |\phi\rangle$ for any single-qubit states.

Operators on Tensor Product Spaces

1. Tensor product of operators:

$$(\hat{A} \otimes \hat{B})(|\psi\rangle \otimes |\phi\rangle) = (\hat{A}|\psi\rangle) \otimes (\hat{B}|\phi\rangle) \quad (85)$$

2. Matrix representation: If $\hat{A}$ is $m \times m$ and $\hat{B}$ is $n \times n$, then $\hat{A} \otimes \hat{B}$ is $mn \times mn$:

$$\hat{A} \otimes \hat{B} = \begin{pmatrix} A_{11}\hat{B} & A_{12}\hat{B} & \cdots \\ A_{21}\hat{B} & A_{22}\hat{B} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \quad (86)$$

Example 15 (Pauli Matrices Tensor Product).

$$\sigma_z \otimes \sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (87)$$

3. Identity operators:

$$\mathbb{I}_1 \otimes \hat{B} \quad \text{acts only on second subsystem} \quad (88)$$

Physical Examples

1. Two-particle systems:

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{particle 1}} \otimes \mathcal{H}_{\text{particle 2}} \quad (89)$$

2. Spin and position:

$$\mathcal{H} = \mathcal{H}_{\text{spatial}} \otimes \mathcal{H}_{\text{spin}} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2 \quad (90)$$

Wavefunction: $\psi_{\uparrow}(\vec{r})$ and $\psi_{\downarrow}(\vec{r})$

3. Multiple qubits:

$$\mathcal{H}_n = \underbrace{\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}_{n \text{ times}} = \mathbb{C}^{2^n} \quad (91)$$

This exponential growth is the basis for quantum computational advantage!

Entanglement

A state $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ is:

- **Separable** if $|\Psi\rangle = |\psi\rangle \otimes |\phi\rangle$ for some $|\psi\rangle \in \mathcal{H}_1, |\phi\rangle \in \mathcal{H}_2$
- **Entangled** otherwise

Entanglement criterion: For a pure state in $\mathbb{C}^2 \otimes \mathbb{C}^2$:

$$|\Psi\rangle = \sum_{i,j=0}^1 c_{ij} |i\rangle |j\rangle \quad (92)$$

The state is separable if and only if the coefficient matrix $C = (c_{ij})$ has rank 1, i.e., $\det(C) = 0$.

Equivalently: $|\Psi\rangle$ is separable iff $c_{00}c_{11} = c_{01}c_{10}$.

Example 16 (Testing for Entanglement). *Separable state:*

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (93)$$

Coefficient matrix: $C = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\det(C) = 0$

Entangled state (Bell state):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (94)$$

Coefficient matrix: $C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\det(C) = \frac{1}{2} \neq 0$

Reduced Density Matrices

For a composite system in state $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$, we define:

Reduced density matrix for subsystem 1:

$$\hat{\rho}_1 = \text{Tr}_2(|\Psi\rangle \langle \Psi|) \quad (95)$$

Reduced density matrix for subsystem 2:

$$\hat{\rho}_2 = \text{Tr}_1(|\Psi\rangle \langle \Psi|) \quad (96)$$

where Tr_i denotes the partial trace over subsystem i .

Entanglement test: A pure state is entangled if and only if its reduced density matrices are mixed (not pure):

$$\text{Tr}(\hat{\rho}_1^2) < 1 \quad \text{or} \quad \text{Tr}(\hat{\rho}_2^2) < 1 \quad (97)$$

Example 17 (Reduced Density Matrix). *For the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:*

$$\hat{\rho}_1 = \text{Tr}_2(|\Phi^+\rangle \langle \Phi^+|) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \mathbb{I} \quad (98)$$

This is a maximally mixed state, confirming maximum entanglement!

$\text{Tr}(\hat{\rho}_1^2) = \frac{1}{2} < 1$, so the state is entangled.

Schmidt Decomposition

Theorem 5 (Schmidt Decomposition). *Any pure state $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ can be written as:*

$$|\Psi\rangle = \sum_{i=1}^r \sqrt{\lambda_i} |u_i\rangle |v_i\rangle \quad (99)$$

where:

- $\{|u_i\rangle\}$ is an orthonormal basis for $\mathcal{H}_1$
- $\{|v_i\rangle\}$ is an orthonormal basis for $\mathcal{H}_2$
- $\lambda_i > 0$ are the Schmidt coefficients with $\sum_i \lambda_i = 1$
- r is the Schmidt rank (number of non-zero coefficients)

Entanglement measure: The state is separable iff $r = 1$, and maximally entangled iff all non-zero λ_i are equal.

Applications and Importance

1. **Quantum computing:** Multi-qubit states require tensor product spaces
2. **Quantum entanglement:** Foundation for quantum information theory
3. **Many-body physics:** Describing systems of multiple particles
4. **Quantum field theory:** Creation/annihilation operators act on tensor product spaces
5. **Composite observables:** Angular momentum addition, spin-orbit coupling

Key insight: The tensor product structure allows quantum systems to exhibit correlations (entanglement) that have no classical analog, forming the basis for quantum computation, quantum cryptography, and quantum teleportation.